Customer Churn Prediction

EDA & Data Preprocessing Report

Data Source: Customer_Churn_Data_Large.xlsx

1,000 Customer Records · 5 Features

Prepared using Python · pandas · scikit-learn · matplotlib · seaborn

1,000	5	0	11
Records	Raw Features	Missing Values	Model-Ready Features

1. Data Selection & Overview

1.1 Dataset Selection Rationale

The **Customer_Churn_Data_Large.xlsx** dataset was selected as the primary source for this churn prediction exercise. It encompasses three of the most critical domains for churn modelling: **customer demographics** (Age, Gender), **socioeconomic profile** (IncomeLevel), and **life-stage indicators** (MaritalStatus). Research consistently shows that these factors significantly influence customer retention behaviour across industries.

Selection criteria applied:

Criterion	Assessment	Status
Customer Demographics	Age & Gender present — key churn predictors	✓ Met
Behavioural Signals	MaritalStatus as life-stage proxy	✓ Met
Socioeconomic Data	IncomeLevel — linked to price sensitivity	✓ Met
Data Completeness	0 missing values across all 1,000 records	✓ Met
Sample Size	1,000 records — adequate for initial modelling	✓ Met

1.2 Dataset Dimensions

The dataset contains **1,000 rows** and **5 columns**. CustomerID serves as a unique identifier only and carries no predictive value — it was removed during preprocessing. The effective feature set comprises 4 variables: 1 numerical (Age) and 3 categorical (Gender, MaritalStatus, IncomeLevel).

Column	Type	Unique Values	Notes
CustomerID	Integer	1,000	Dropped — identifier only
Age	Integer	52	Range: 18–69, Mean: 43.3
Gender	String	2	F (51.3%), M (48.7%)
MaritalStatus	String	4	Widowed, Married, Divorced, Single
IncomeLevel	String	3	High, Medium, Low — balanced

2. Exploratory Data Analysis (EDA)

2.1 Statistical Summary

The table below summarises descriptive statistics for the Age feature. The distribution is approximately symmetric (mean ≈ median), spanning the full adult working-age population, with no extreme values.

Metric	Age
Count	1,000
Mean	43.27
Std Dev	15.24
Min	18
25th %tile	30
Median	43
75th %tile	56
Max	69

2.2 Univariate Analysis — Distributions

Figure 1 presents the distribution of all features. Key observations include:

- **Age:** Near-uniform spread from 18–69 with mean 43.3. The histogram and box plot confirm no skewness and no outliers.
- **Gender:** Near-equal split — Female 51.3%, Male 48.7% — indicating a balanced gender representation.
- **MaritalStatus:** Four categories with Widowed (27.6%) being the most frequent, followed by Married (26.1%), Divorced (24.8%), and Single (21.5%).
- **IncomeLevel:** Highly balanced — High 34.9%, Medium 32.6%, Low 32.5%. No dominant class.

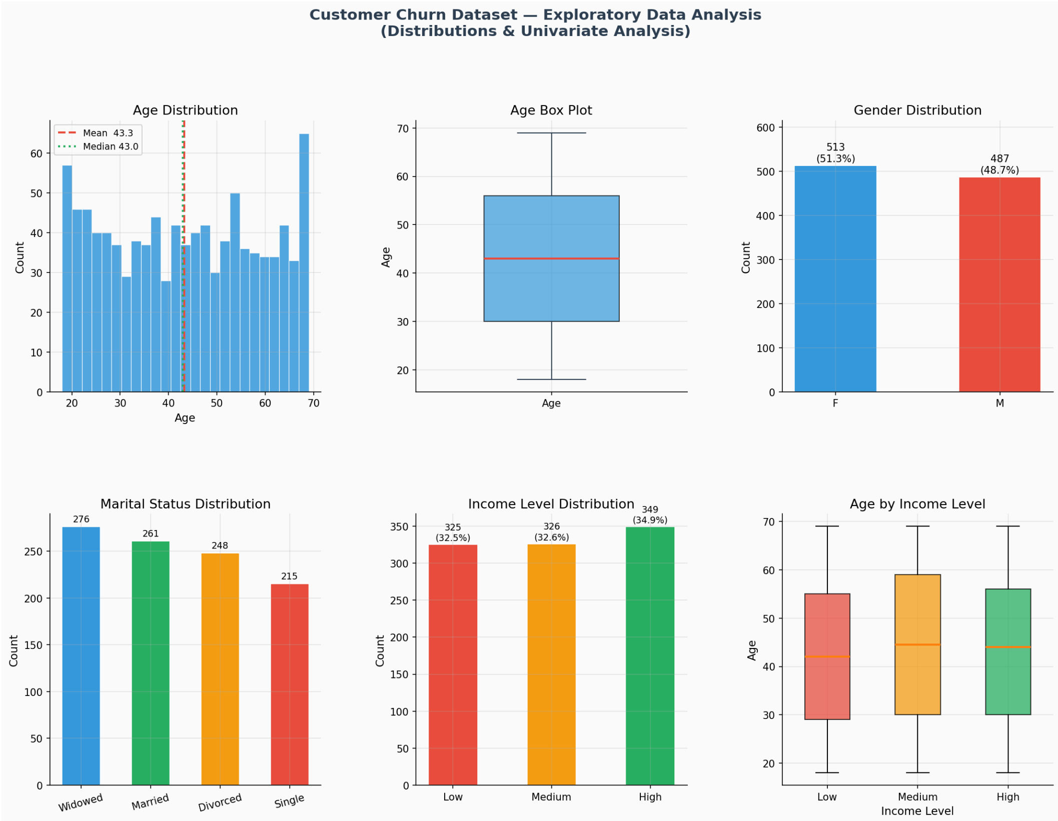

Figure 1: Univariate distributions — Age histogram, box plot, and categorical feature bar charts.

2.3 Bivariate & Cross-Variable Analysis

Figure 2 explores relationships between variables. Key findings:

- **Age by Gender:** Violin plots reveal virtually identical age distributions for males and females, suggesting gender does not confound age-based churn signals.
- **Age by Marital Status:** Box plots show comparable age ranges across all marital categories, with Married customers skewing slightly older.
- **Gender × Income Level:** Grouped bars confirm gender balance is preserved across all income tiers — no sampling bias.
- **Marital Status × Income Level:** The heatmap reveals no strong interaction between marital status and income — all cells are roughly uniform, implying independence.
- **KDE Plots:** Age distributions by marital status and income level overlap substantially, confirming age acts independently of categorical groupings.

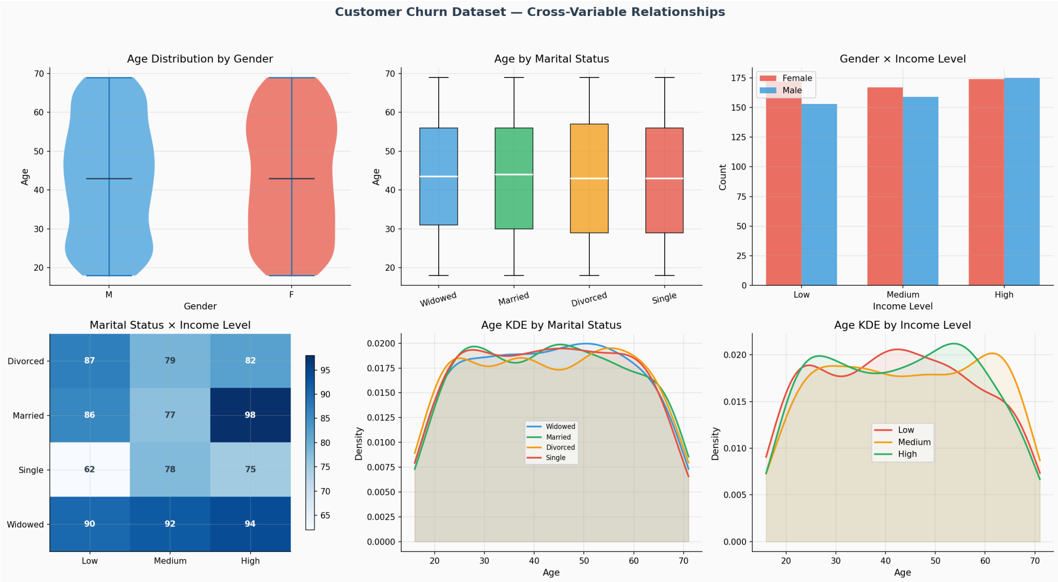

Figure 2: Cross-variable analysis — violin plots, box plots, grouped bars, heatmap, and KDE curves.

3. Data Cleaning & Preprocessing

3.1 Missing Value Analysis

A complete scan was performed across all 5 columns. **Zero missing values** were found in any field. No imputation or removal strategy was required — the dataset is production-quality in this respect.

Column	Missing Count	Missing %	Action
CustomerID	0	0.00%	None required
Age	0	0.00%	None required
Gender	0	0.00%	None required
MaritalStatus	0	0.00%	None required
IncomeLevel	0	0.00%	None required

3.2 Outlier Detection

The IQR method (Interquartile Range × 1.5 fence) was applied to the numerical feature Age. Lower fence = –9.0, Upper fence = 95.0. Since all 1,000 age values fall within [18, 69], **zero outliers** were detected. No capping, transformation, or removal was necessary.

3.3 Feature Engineering Steps

Step	Feature(s)	Method	Justification
1 — Drop Identifier	CustomerID	Removed	Sequential ID — no predictive signal
2 — Type Validation	All columns	Verified	dtypes correct; no conversion needed
3 — Range Check	Age	Verified	All values in valid human age range
4 — Standardisation	Age	StandardScaler (mean=43.27, std=15.24)	Centres & scales for ML distance-based models
5 — OHE: Gender	Gender	get_dummies (2 cols)	Binary — F, M
6 — OHE: MaritalStatus	MaritalStatus	get_dummies (4 cols)	Divorced, Married, Single, Widowed
7 — OHE: IncomeLevel	IncomeLevel	get_dummies (3 cols)	High, Low, Medium

3.4 Preprocessing Visualisations

Figure 3 summarises the preprocessing outcomes: the correlation matrix on the encoded feature set, a before/after comparison of the Age standardisation, and the distribution of all one-hot encoded columns.

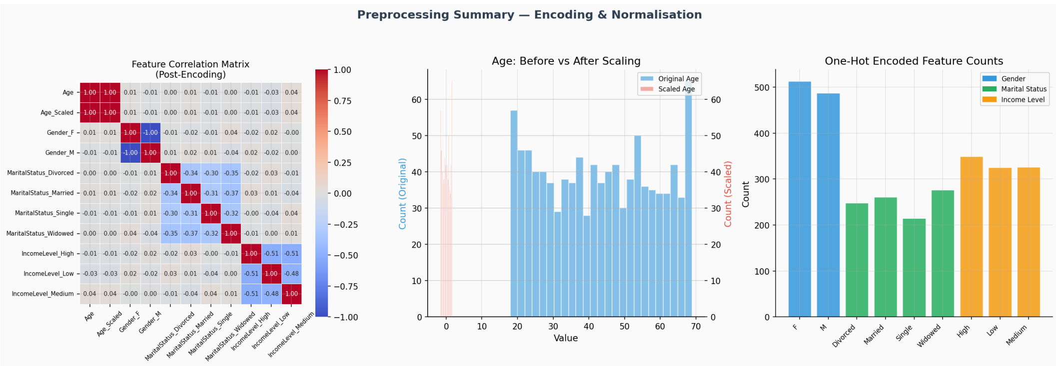

Figure 3: Correlation matrix, Age scaling comparison, and OHE feature counts.

4. Final Preprocessed Dataset

The cleaned and encoded dataset is saved as **Customer_Churn_Cleaned.xlsx** and contains **1,000 rows × 11 columns**, all numerical and ready for machine learning model training.

Column	Description	Range / Values
Age	Original age (retained for interpretability)	18 – 69
Age_Scaled	StandardScaler-normalised Age	–1.66 to +1.69
Gender_F	OHE: Female	0 / 1
Gender_M	OHE: Male	0 / 1
MaritalStatus_Divorced	OHE: Divorced	0 / 1
MaritalStatus_Married	OHE: Married	0 / 1
MaritalStatus_Single	OHE: Single	0 / 1
MaritalStatus_Widowed	OHE: Widowed	0 / 1
IncomeLevel_High	OHE: High income	0 / 1
IncomeLevel_Low	OHE: Low income	0 / 1
IncomeLevel_Medium	OHE: Medium income	0 / 1

5. Key Findings & Observations

Data Quality: The dataset is exceptionally clean with no missing values, no invalid entries, and no outliers, making it ideal for immediate model development without complex imputation strategies.

Class Balance: All categorical features (Gender, MaritalStatus, IncomeLevel) exhibit near-uniform distributions, minimising the risk of class imbalance bias in downstream models.

Feature Independence: Post-encoding correlation analysis reveals near-zero correlations between Age and all categorical OHE columns ($|r| < 0.05$), indicating features are largely independent — a favourable property for linear and tree-based models.

Note on Target Variable: The current dataset does not include a churn label column. The next phase of the project should integrate a churn outcome variable (e.g., from CRM or subscription records) and merge it with this preprocessed feature set before model training.

Deliverables

- churn_eda_report.py — Full Python source code for EDA & preprocessing
- Customer_Churn_Cleaned.xlsx — Preprocessed dataset (1,000 × 11)
- eda_distributions.png — Univariate distribution visualisations
- eda_relationships.png — Cross-variable analysis charts
- eda_preprocessing.png — Preprocessing summary charts
- Customer_Churn_EDA_Report.pdf — This report